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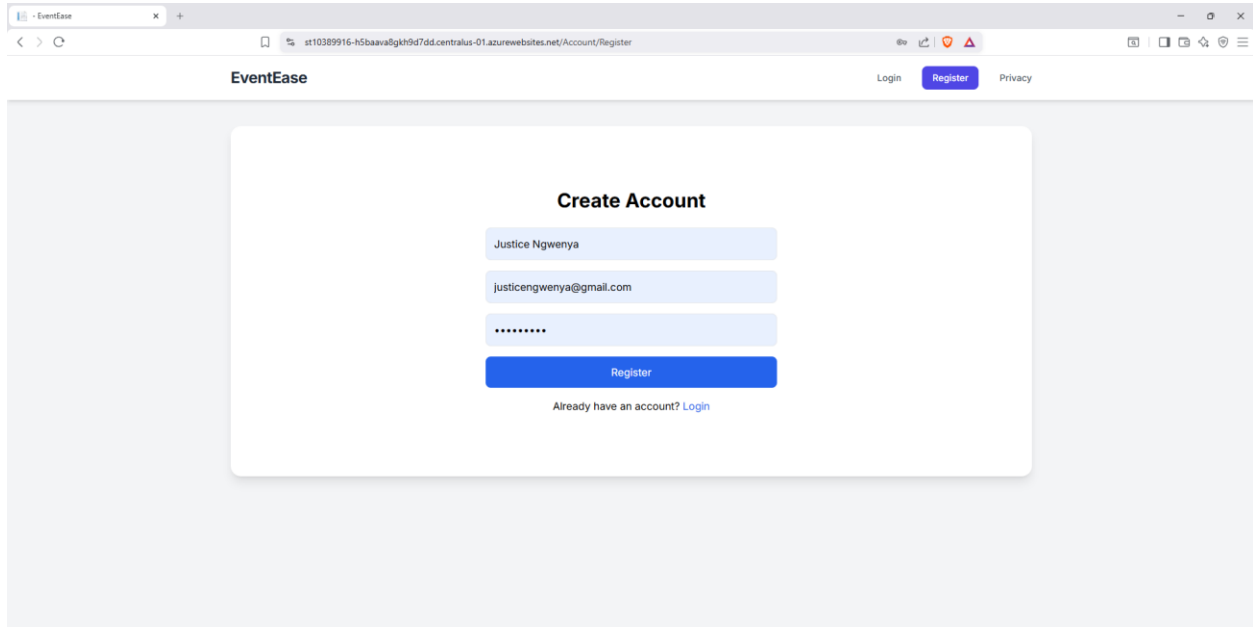
Justice Ngwenya
ST10389916

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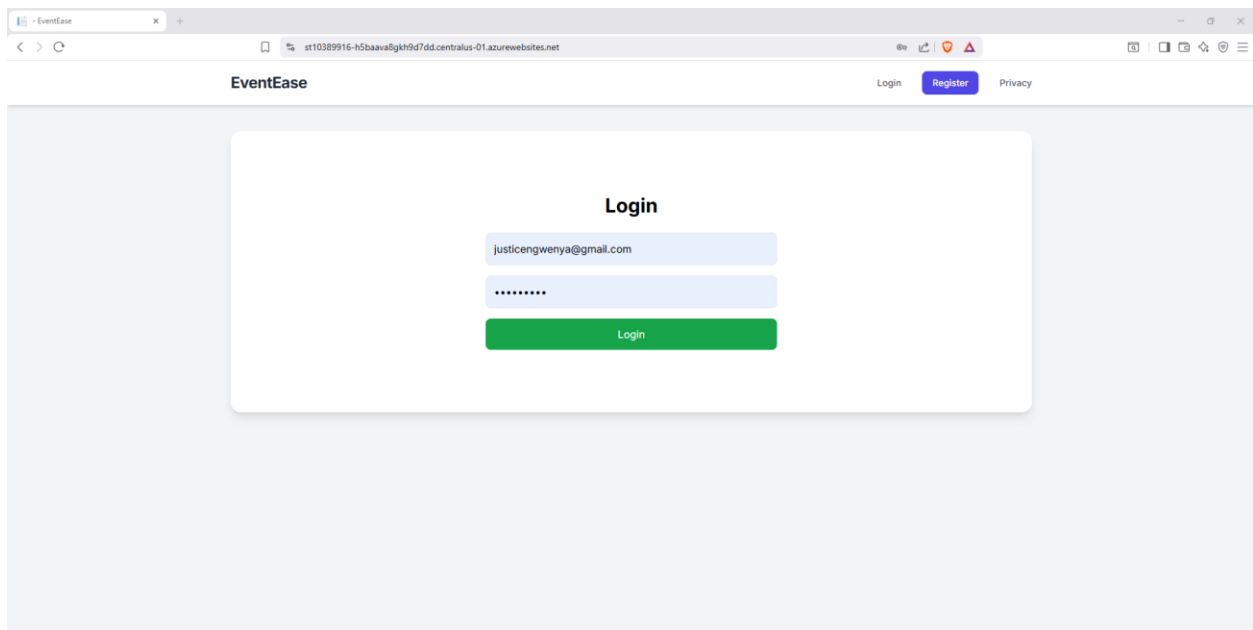
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Event Ease Application

Registration and Login

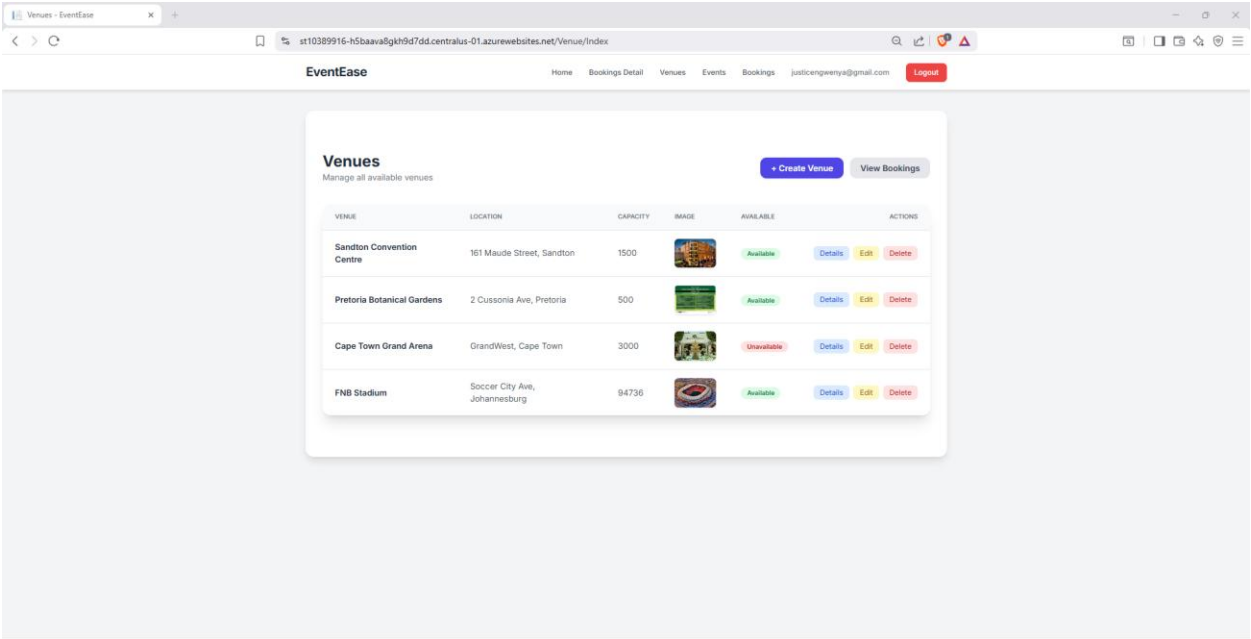


The screenshot shows the registration page of the EventEase application. The browser's address bar displays the URL: `st10389916-h5baava8gh9d7dd.centralus-01.azurewebsites.net/Account/Register`. The page features a header with the "EventEase" logo, a "Login" link, a "Register" button, and a "Privacy" link. The main content area contains a "Create Account" form with the following fields: a text input for the name "Justice Ngwenya", an email input for "justicengwenya@gmail.com", and a password input represented by dots. A blue "Register" button is positioned below these fields. At the bottom of the form, there is a link that says "Already have an account? Login". The footer of the page includes the copyright notice: "© 2026 - EventEase. All rights reserved."

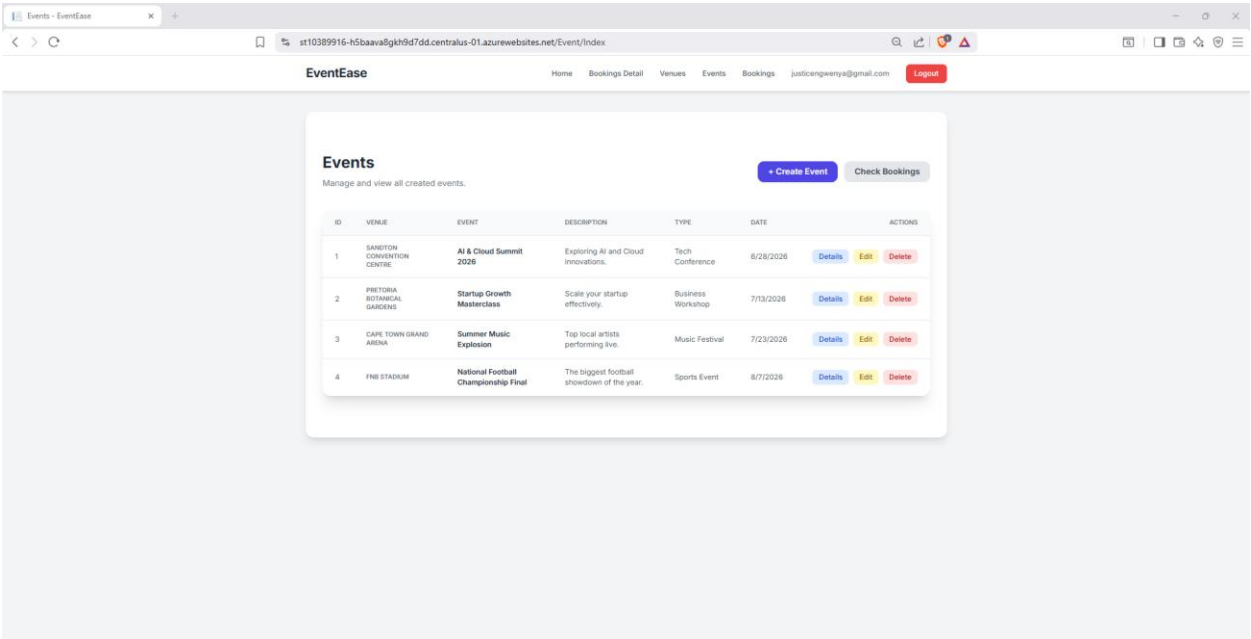


The screenshot shows the login page of the EventEase application. The browser's address bar displays the URL: `st10389916-h5baava8gh9d7dd.centralus-01.azurewebsites.net`. The page features a header with the "EventEase" logo, a "Login" link, a "Register" button, and a "Privacy" link. The main content area contains a "Login" form with the following fields: an email input for "justicengwenya@gmail.com" and a password input represented by dots. A green "Login" button is positioned below these fields. The footer of the page includes the copyright notice: "© 2026 - EventEase. All rights reserved."

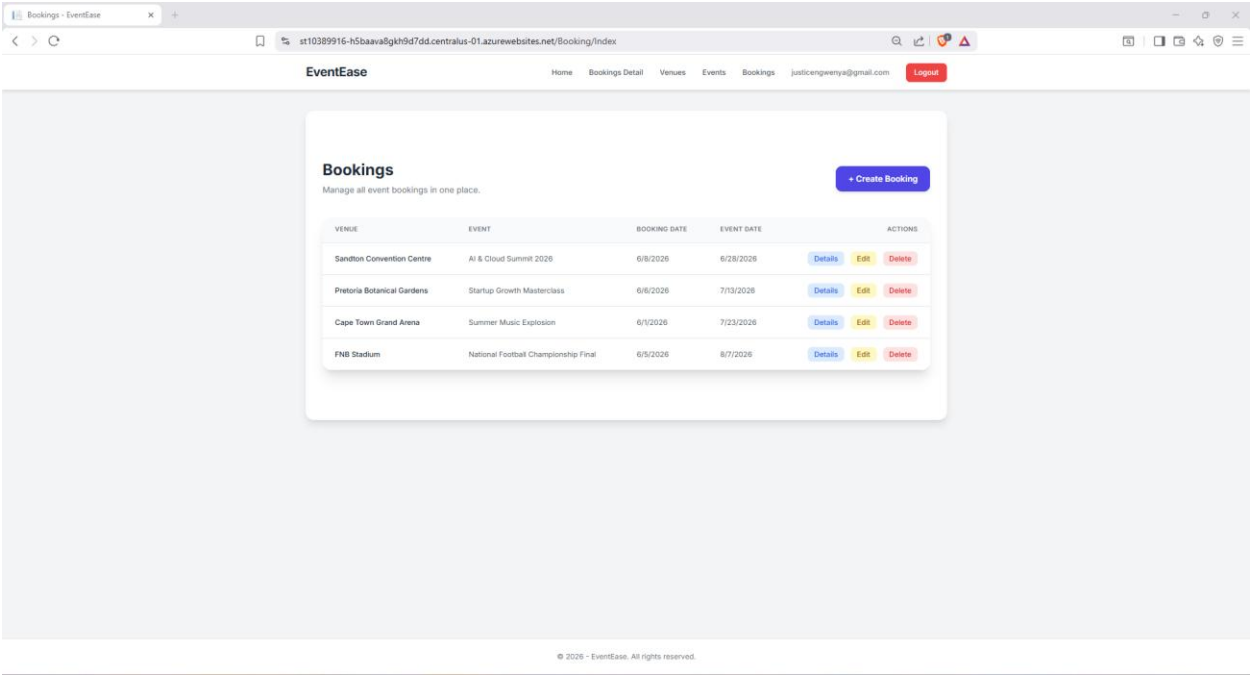
Venues



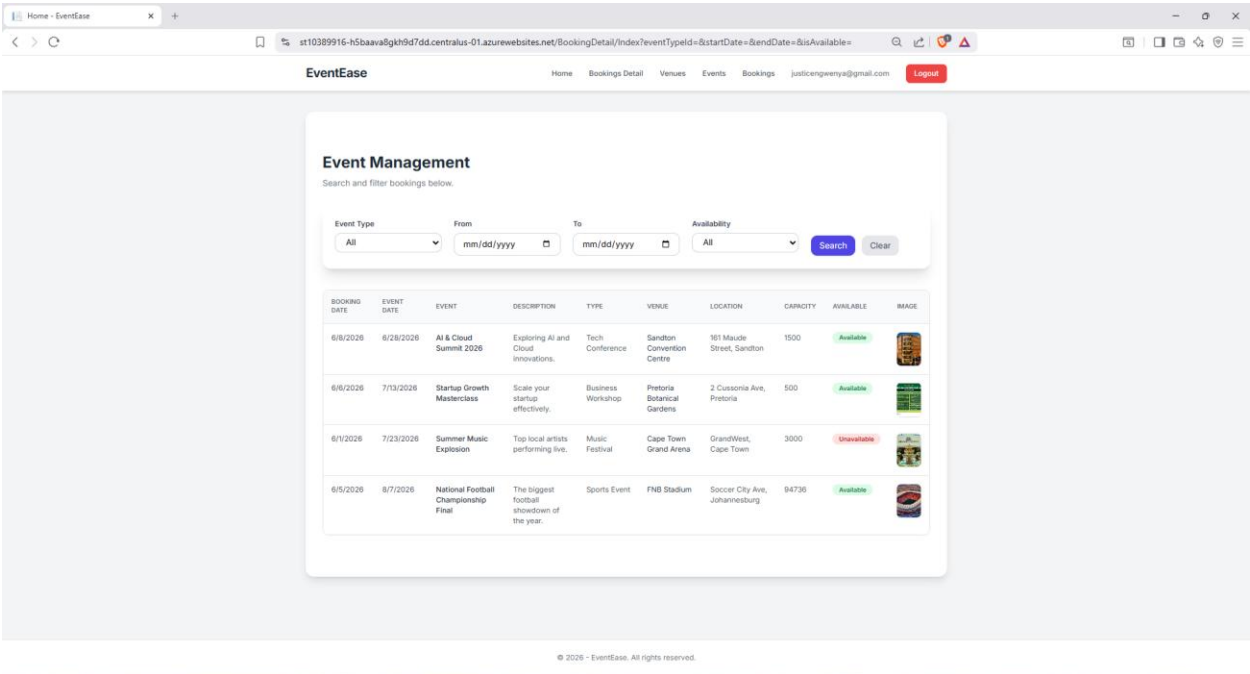
Events



Bookings



Advanced Filtering



Filter bookings via EventType

The screenshot shows the EventEase application interface. The header includes the logo 'EventEase' and navigation links: Home, Bookings Detail, Venues, Events, Bookings, and a user profile 'justicengwenya@gmail.com' with a 'Logout' button. The main content area is titled 'Event Management' with the subtitle 'Search and filter bookings below.' Below this is a search filter bar with four dropdown menus: 'Event Type' (set to 'Tech Conference'), 'From' (date picker), 'To' (date picker), and 'Availability' (set to 'All'). A 'Search' button and a 'Clear' button are also present. Below the filter bar is a table of bookings. The table has columns: Booking Date, Event Date, Event, Description, Type, Venue, Location, Capacity, Available, and Image. One booking is visible: 'AI & Cloud Summit 2026' on '6/8/2026' at 'Sandton Convention Centre', with a capacity of 1500 and status 'Available'.

BOOKING DATE	EVENT DATE	EVENT	DESCRIPTION	TYPE	VENUE	LOCATION	CAPACITY	AVAILABLE	IMAGE
6/8/2026	6/28/2026	AI & Cloud Summit 2026	Exploring AI and Cloud innovations.	Tech Conference	Sandton Convention Centre	161 Maule Street, Sandton	1500	Available	

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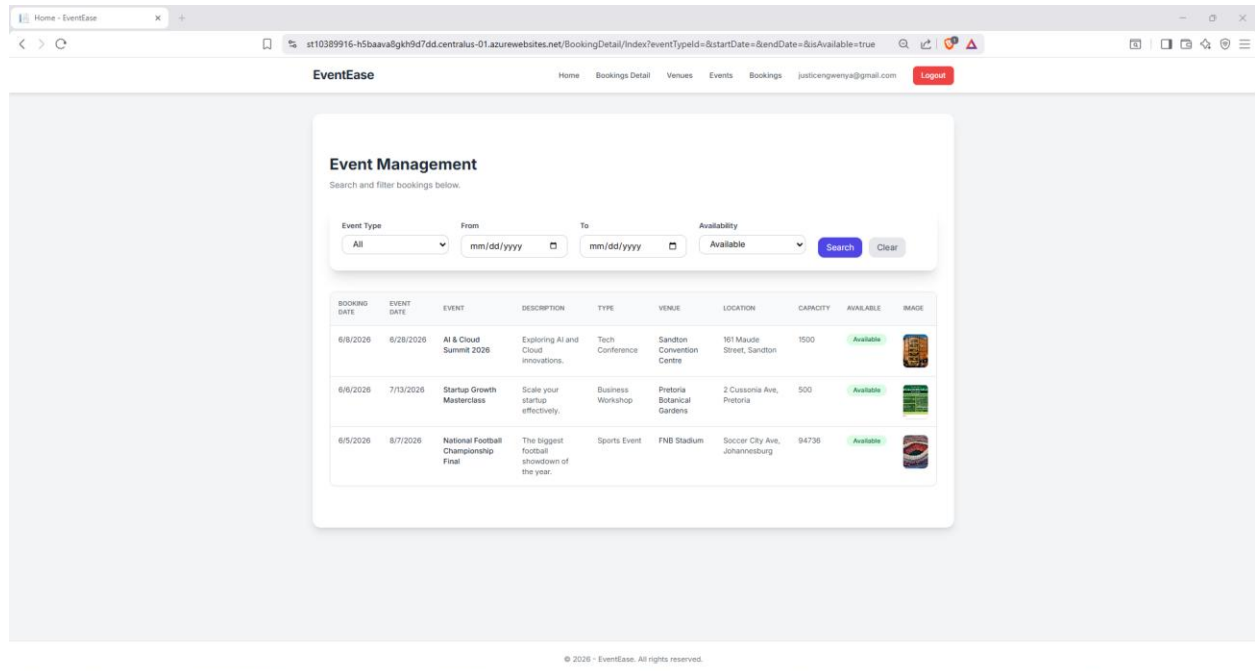
Filter bookings on a date range (06/01/2026 – 07/31/2026)

The screenshot shows the EventEase application interface with the search filter bar updated. The 'Event Type' dropdown is now set to 'All'. The 'From' date picker is set to '06/01/2026' and the 'To' date picker is set to '07/31/2026'. The 'Availability' dropdown remains set to 'All'. The 'Search' button is highlighted. Below the filter bar, the table of bookings is updated to show three events within the specified date range: 'AI & Cloud Summit 2026', 'Startup Growth Masterclass', and 'Summer Music Explosion'.

BOOKING DATE	EVENT DATE	EVENT	DESCRIPTION	TYPE	VENUE	LOCATION	CAPACITY	AVAILABLE	IMAGE
6/8/2026	6/28/2026	AI & Cloud Summit 2026	Exploring AI and Cloud innovations.	Tech Conference	Sandton Convention Centre	161 Maule Street, Sandton	1500	Available	
6/6/2026	7/13/2026	Startup Growth Masterclass	Scale your startup effectively.	Business Workshop	Pretoria Botanical Gardens	2 Cussonia Ave, Pretoria	500	Available	
6/9/2026	7/23/2026	Summer Music Explosion	Top local artists performing live.	Music Festival	Cape Town Grand Arena	Grandditch, Cape Town	3000	Unavailable	

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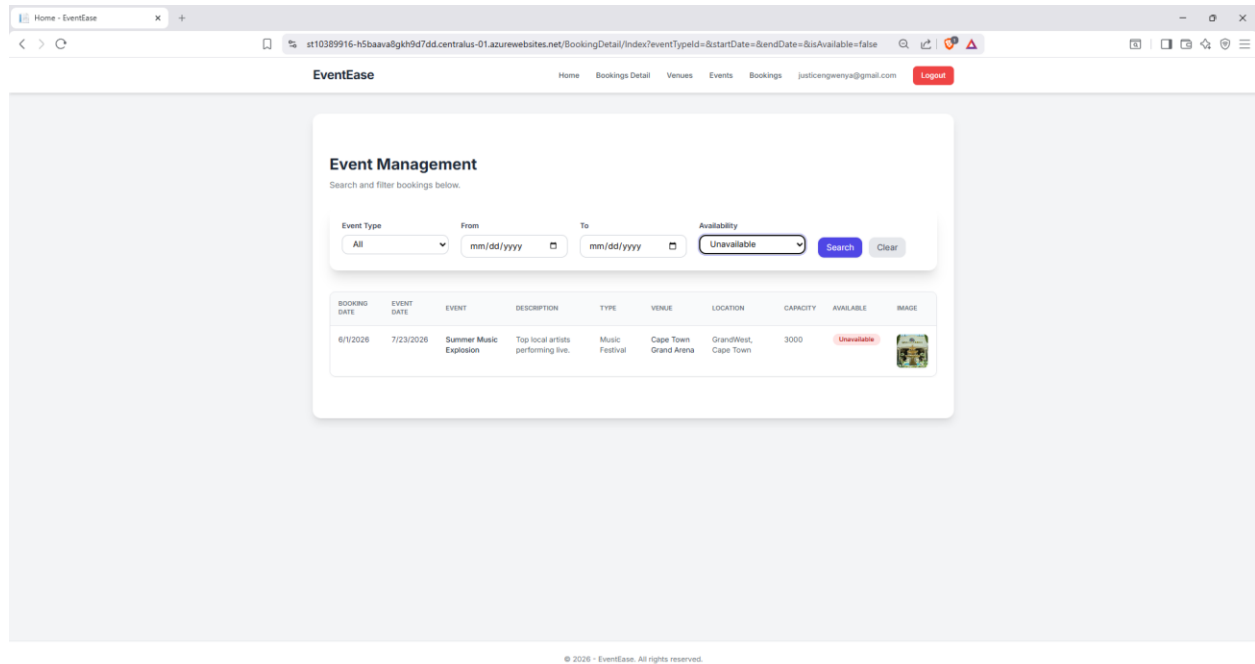
Filter bookings on Venue Availability



The screenshot shows the EventEase web application. The browser address bar displays the URL: `st10389916-h5baava8gh9d7dd.centralus-01.azurewebsites.net/BookingDetail/Index?eventTypeId=&startDate=&endDate=&isAvailable=true`. The navigation bar includes links for Home, Bookings Detail, Venues, Events, Bookings, and a Logout button. The main content area is titled "Event Management" with the subtitle "Search and filter bookings below." Below this is a filter bar with fields for Event Type (set to "All"), From date (mm/dd/yyyy), To date (mm/dd/yyyy), and Availability (set to "Available"). A "Search" button and a "Clear" button are also present. The search results are displayed in a table with the following data:

BOOKING DATE	EVENT DATE	EVENT	DESCRIPTION	TYPE	VENUE	LOCATION	CAPACITY	AVAILABLE	IMAGE
6/8/2026	6/28/2026	AI & Cloud Summit 2026	Exploring AI and Cloud innovations.	Tech Conference	Sandton Convention Centre	161 Maude Street, Sandton	1500	Available	
6/6/2026	7/13/2026	Startup Growth Masterclass	Scale your startup effectively.	Business Workshop	Pretoria Botanical Gardens	2 Cussonia Ave, Pretoria	500	Available	
6/5/2026	8/7/2026	National Football Championship Final	The biggest football showdown of the year.	Sports Event	FNB Stadium	Soccer City Ave, Johannesburg	94736	Available	

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The screenshot shows the EventEase web application with the same URL as the first image. The filter bar now has the Availability dropdown set to "Unavailable". The search results table contains only one entry:

BOOKING DATE	EVENT DATE	EVENT	DESCRIPTION	TYPE	VENUE	LOCATION	CAPACITY	AVAILABLE	IMAGE
6/1/2026	7/23/2026	Summer Music Explosion	Top local artists performing live.	Music Festival	Cape Town Grand Arena	GrandWest, Cape Town	3000	Unavailable	

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Code base GitHub Link

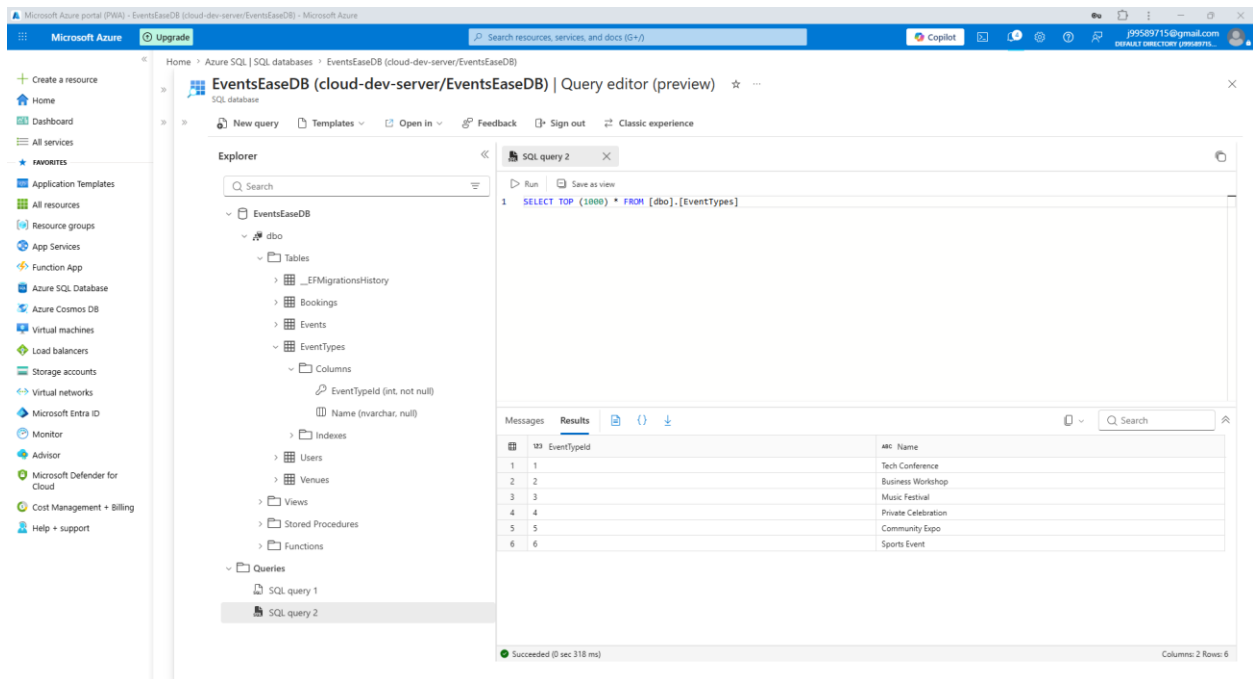
<https://github.com/ST10389916/EventEase>

Azure Deployment Updates

Web app URL

<https://st10389916-h5baava8gkh9d7dd.centralus-01.azurewebsites.net/>

EventTypes Table



Microsoft Azure portal (PWA) - EventsEaseDB (cloud-dev-server/EventsEaseDB) - Microsoft Azure

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Azure SQL > SQL databases > EventsEaseDB (cloud-dev-server/EventsEaseDB)

EventsEaseDB (cloud-dev-server/EventsEaseDB) | Query editor (preview)

SQL database

New query Templates Open in Feedback Sign out Classic experience

SQL query 2

Run Save as view

1 SELECT TOP (1000) * FROM [dbo].[EventTypes]

Explorer

Search

EventsEaseDB

dbo

Tables

_EFMigrationsHistory

Bookings

Events

EventTypes

Columns

EventTypeId (int, not null)

Name (nvarchar, null)

Indexes

Users

Venues

Views

Stored Procedures

Functions

Queries

SQL query 1

SQL query 2

Messages Results ()

EventTypeId AR: Name

1	1	Tech Conference
2	2	Business Workshop
3	3	Music Festival
4	4	Private Celebration
5	5	Community Expo
6	6	Sports Event

Succeeded (0 sec 318 ms) Columns: 2 Rows: 6

Microsoft Azure portal (PWA) - EventsEaseDB (cloud-dev-server/EventsEaseDB) - Microsoft Azure

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > EventsEaseDB (cloud-dev-server/EventsEaseDB) | Query editor (preview)

SQL database

New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance Monitoring Automation Help

Explorer

EventsEaseDB

dbo

Tables

__EFMigrations Bookings Events EventTypes Users Venues

Views Stored Procedures Functions

Queries

SQL query 1

SQL query 1

Run Save as view

1 SELECT TOP (1000) * FROM [dbo].[EventTypes]

Messages Results Messages Results Messages Results Messages Results

103	EventTypeId	440	Name
1	1		Tech Conference
2	2		Business Workshop
3	3		Music Festival
4	4		Private Celebration
5	5		Community Expo
6	6		Sports Event

Succeeded (0 sec 353 ms) Columns: 2 Rows: 6

Microsoft Azure portal (PWA) - EventsEaseDB (cloud-dev-server/EventsEaseDB) - Microsoft Azure

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > EventsEaseDB (cloud-dev-server/EventsEaseDB) | Query editor (preview)

SQL database

New query Templates Open in Feedback Sign out Classic experience

Explorer

EventsEaseDB

dbo

Tables

__EFMigrations Bookings Events EventTypes Users Venues

Views Stored Procedures Functions

SQL query 1

Venues

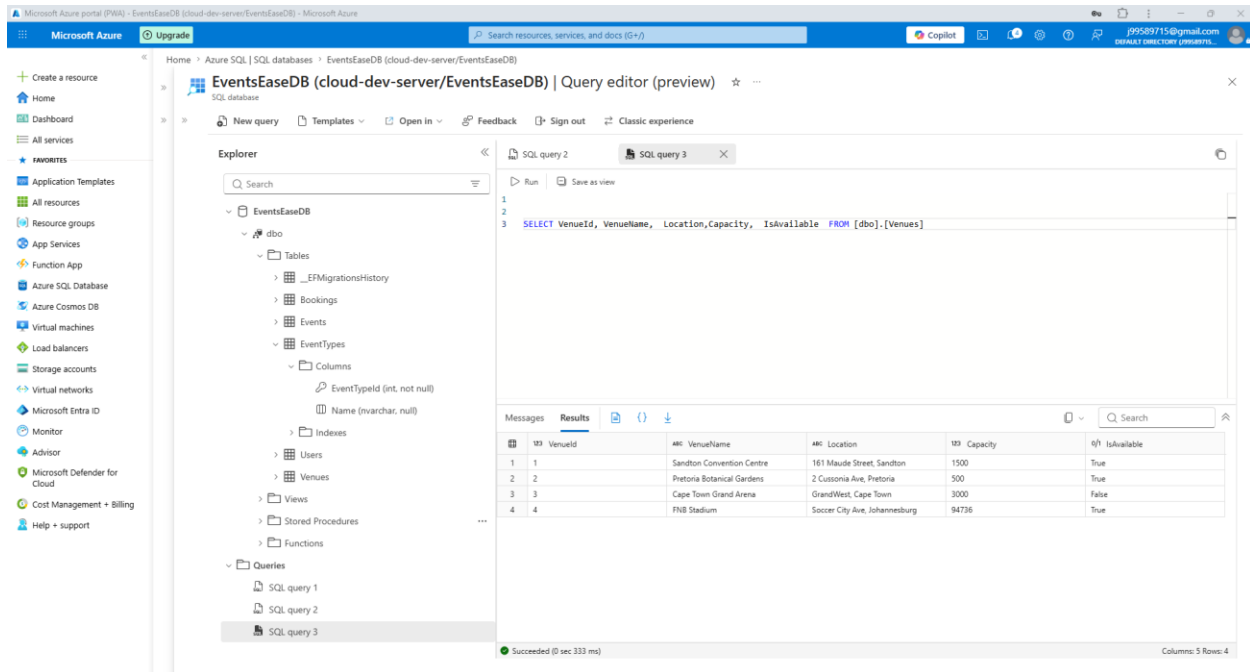
Events

Data preview - Events

Showing 1000 rows

103	EventId	103	EventTypeId	103	VenueId	440	EventName	440	Description	6	EventDate
1	1		1		1		AI & Cloud Summit 2026		Exploring AI and Cloud innovations.		2026-06-28T00:00:00.0000000
2	2		2		2		Startup Growth Masterclass		Scale your startup effectively.		2026-07-13T00:00:00.0000000
3	3		3		3		Summer Music Explosion		Top local artists performing live.		2026-07-23T00:00:00.0000000
4	4		6		4		National Football Championship FL...		The biggest football showdown of...		2026-08-07T00:00:00.0000000

Succeeded (0 sec 326 ms) Columns: 6 Rows: 4



Reflective Technical Report

Introduction

The EventEase project was developed as a cloud-based event management platform intended to streamline the administration of venues, events, bookings, and users through a centralized online system. The main objective of the application was to provide administrators with a secure and efficient environment for managing event-related activities while demonstrating the practical application of modern cloud computing technologies. The system was created using ASP.NET Core MVC and deployed through Microsoft Azure services, specifically Azure App Service and Azure SQL Database. By integrating cloud hosting, relational database management, authentication, and responsive web design, the project resulted in a fully operational web-based solution.

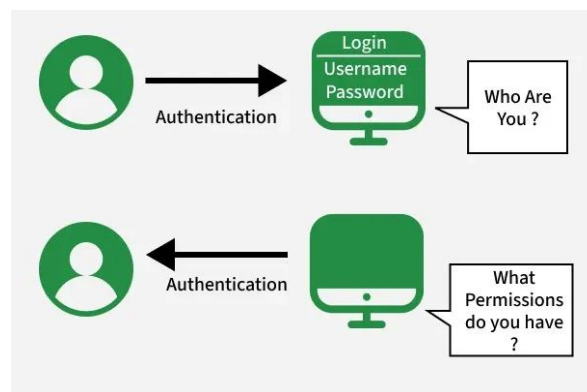
The development process of EventEase offered valuable practical experience in several areas of software engineering, including cloud deployment, database design, scalability, security, and application architecture. Throughout the implementation process, various cloud services and development technologies were analyzed and incorporated to ensure the application remained scalable, reliable, and easy to maintain. In addition, the project created an opportunity to evaluate the differences between traditional on-premises

systems and cloud-hosted solutions, particularly highlighting why Platform as a Service (PaaS) was the most appropriate deployment model for this application (scielo, 2023).

Detailed Description of the Application's Full Feature List

The EventEase system includes several integrated features designed to support efficient event management operations. The application was structured using the MVC architectural pattern alongside a relational database model to ensure organized data management and maintainable system development. Each feature was carefully implemented to address specific operational requirements while contributing to the overall effectiveness of the platform.

User Authentication and Authorization



The EventEase system includes a secure authentication and authorization module that allows users to register, log in, and access the platform safely. It ensures that only authorized users can manage venues, events, and bookings, protecting sensitive information from unauthorized access. Implemented using ASP.NET Core MVC, the module incorporates secure user management practices, session handling, user verification, password protection, and restricted administrator access, with administrative accounts configured for testing and demonstration purposes (microsoft, 2025).

Venue Management Module

The Venue Management Module enables administrators to efficiently manage venue information within the EventEase system. It supports the creation, viewing, updating, and deletion of venue records, including details such as venue name, location, capacity, availability status, and images. The module provides a centralized repository for venues that can be linked to events and bookings, while demonstrating the integration of image management and relational database functionality within a cloud-hosted environment. Complete CRUD operations ensure flexible and effective venue administration.

Event Management Module

The Event Management Module is a core component of the EventEase system that allows administrators to create, manage, and organize events linked to specific venues and event categories. It supports complete CRUD functionality, enabling efficient creation, viewing, updating, and deletion of event records. Event information includes names, descriptions, dates, event types, and venue associations. Through the use of foreign keys and relational database structures, the module maintains strong relationships between events, venues, and event types, ensuring data integrity and consistency throughout the application.

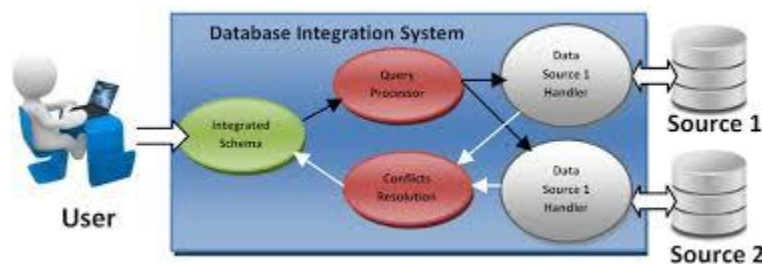
Booking Management Module

The Booking Management Module handles reservation and scheduling activities within the EventEase system. It allows administrators to create, manage, and monitor bookings associated with specific events and venues. The module supports full CRUD functionality, enabling users to add, update, view, and delete booking records efficiently. Booking information includes reservation dates, event associations, venue details, and scheduling data. Through the use of relational database structures and entity relationships, the module ensures accurate data management and seamless integration within the cloud-based application environment.

Responsive User Interface

The EventEase application features a responsive and user-friendly interface developed using ASP.NET MVC Views. Designed to enhance usability and simplify navigation, the interface provides administrators with easy access to key system functions through landing pages, dashboards, management pages, login screens, and informational pages. The use of forms, tables, navigation menus, and detail views improves accessibility and data presentation, while the structured layout ensures a professional appearance and an intuitive user experience across the platform (eleken, 2025).

Database Integration



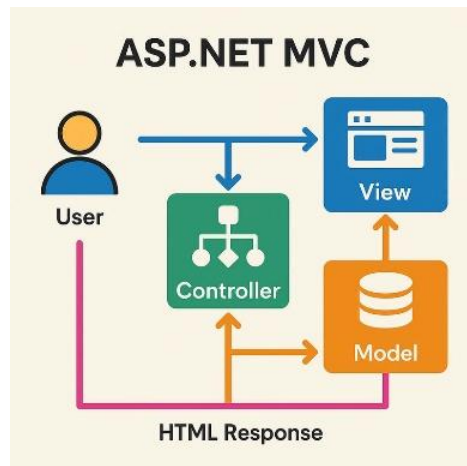
Database integration is a fundamental component of the EventEase system, utilizing an Azure SQL Database to store and manage application data, including users, venues, events, event types, and bookings. The database employs a relational structure with primary and foreign keys to maintain data integrity, consistency, and accurate relationships between entities. Supported by an Entity Relationship Diagram (ERD), the design enables efficient data storage, retrieval, and management while demonstrating effective cloud-based database implementation (kellyblaire, 2024).

Cloud Deployment Features

The EventEase application was deployed on Microsoft Azure App Service, providing a secure cloud-hosted environment with direct integration to Azure SQL Database. This deployment enables the system to be publicly accessible online while benefiting from centralized hosting, scalability, simplified infrastructure management, and remote access capabilities. By leveraging cloud technologies, the platform demonstrates the advantages of modern cloud computing, including secure HTTPS communication, reliable database connectivity, and flexible infrastructure that supports efficient system operation and accessibility (amazon, 2025).

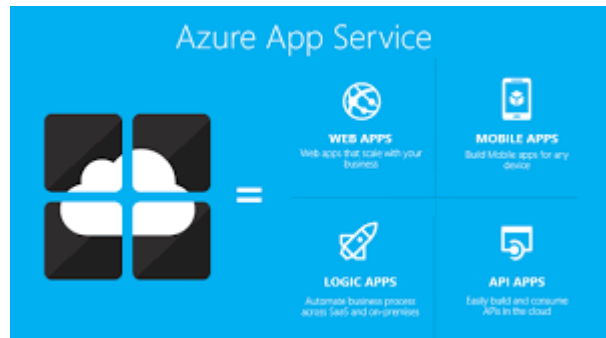
Component Discussion

ASP.NET Core MVC



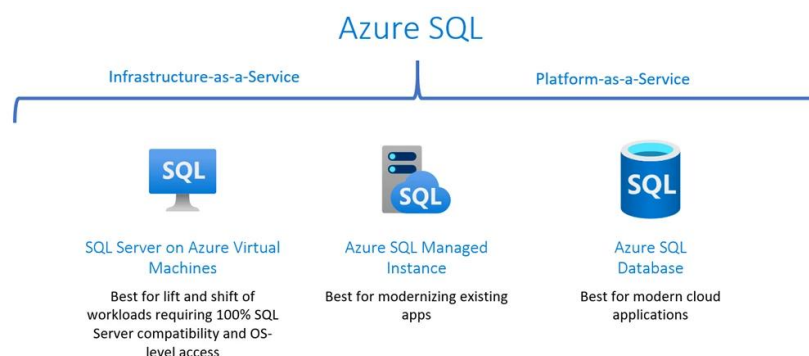
ASP.NET Core MVC was selected as the primary development framework for the EventEase project due to its scalability, maintainability, security, and seamless integration with Microsoft Azure. The MVC architecture separates the application into Models, Views, and Controllers, promoting a clear separation of responsibilities and improving code organization, testing, debugging, and future enhancements. Controllers manage user interactions and application flow, Models handle business logic and data processing, while Views provide the user interface. Its strong support for authentication, database integration, and cloud deployment made it the most suitable choice compared to alternatives such as Django, Laravel, Node.js with Express, and Spring Boot (linkedin, 2022).

Azure App Service



Azure App Service was chosen as the hosting platform for the EventEase system due to its ability to simplify cloud deployment and infrastructure management. As a Platform as a Service (PaaS) solution, it removes the need for manual server, operating system, and network configuration, allowing development efforts to focus on application functionality. The platform provides built-in HTTPS support, automatic scaling, deployment integration, and centralized management, enhancing the system's reliability, security, and accessibility. Compared to alternatives such as AWS Elastic Beanstalk, Google App Engine, Docker container hosting, and traditional IIS hosting, Azure App Service was the most suitable option because of its seamless integration with ASP.NET Core and Azure SQL Database (intercept.cloud, 2025).

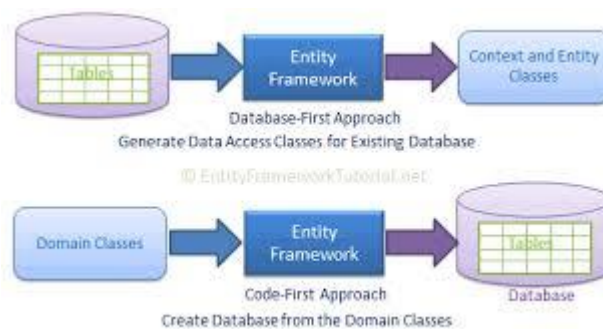
Azure SQL Database



Azure SQL Database was selected as the primary database solution for the EventEase system to manage and store relational data in a secure cloud environment. The platform provides scalability, automated backups, advanced security features, performance monitoring, and reduced administrative overhead. It supports the centralized management of users, events, venues, and bookings while integrating seamlessly with Visual Studio and

Entity Framework Core, simplifying development and deployment. Compared to alternatives such as MySQL, PostgreSQL, MongoDB, and Firebase Firestore, Azure SQL Database was chosen for its strong compatibility with the Microsoft ecosystem and its enterprise-grade reliability and performance (quizlet, 2024).

Entity Framework Core



Entity Framework Core was used as the Object Relational Mapping (ORM) framework to facilitate communication between the EventEase application and the Azure SQL Database. By representing database entities as C# objects, it reduced the need for extensive manual SQL queries, simplifying database operations and improving development efficiency. The framework also streamlined database migrations, schema management, and data access while integrating seamlessly with ASP.NET Core. Compared to alternatives such as raw SQL queries and Dapper ORM, Entity Framework Core provided the optimal balance of productivity, simplicity, and strong framework integration for the project (medium, 2025).

Reflection on the Project

The EventEase project significantly enhanced my understanding of cloud computing and modern web application development by providing practical experience with Microsoft Azure, ASP.NET Core MVC, and cloud-based deployment. Through designing, developing, and deploying the application, I gained hands-on knowledge of cloud services, relational database management, authentication systems, and scalable web application architecture.

A key lesson learned was the importance of proper system planning and database design. Creating the relational database structure and Entity Relationship Diagram (ERD) reinforced the need for well-organized and scalable data models to ensure data integrity, consistency, and efficient system performance. The project also highlighted the value of effective troubleshooting and problem-solving when addressing technical challenges such as Azure SQL Database connectivity issues, authentication and firewall configurations, deployment errors, runtime compatibility problems, and dependency management (researchgate, 2025).

The project further expanded my understanding of cloud security principles, particularly the shared responsibility model, where cloud providers secure the underlying infrastructure while developers are responsible for application-level security. Additionally, I gained insight into the benefits of cloud scalability, managed infrastructure, deployment automation, and Platform as a Service (PaaS) solutions, which reduce operational complexity and improve development efficiency (orca.security, 2025).

Overall, the project strengthened my skills in cloud-native application development, Azure deployment workflows, relational database integration, authentication and authorization, and scalable system design. It also improved my analytical thinking, troubleshooting abilities, and practical software development experience, providing valuable knowledge that will support future academic and professional opportunities.

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